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Unveiling green digital transformational leadership: Nexus between green digital culture, green digital mindset, and green digital transformation

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ABSTRACT

The study's primary purpose has been to examine the association between Green Digital Transformational Leadership, Green Digital Mindset, and Green Digital Transformation Under Varying Permutations of Organizational Green Digital Culture. Using the Stimuli-Organism-Response theory and the transformational leadership theory as the underpinning theoretical frameworks, the direct, indirect, and moderated roles of in effectuating have been investigated. After performing PLS-based structural equation modeling on the data collected from 240 respondents approached via LinkedIn, a direct as well as indirect positive linkage between Green Digital Transformational Leadership, Green Digital Mindset, and Green Digital Transformation has been empirically substantiated, emphasizing the importance of Green Digital Transformational Leadership in enabling Green Digital Transformation through the cultivation of an ecologically aware mindset among their followers. The significant positive moderation effect of Organizational Green Digital Culture on the Green Digital Transformational Leadership - Green Digital Mindset relationship reveals that the nurturing of an effective Organizational Green Digital Culture could amplify the instrumentality of Green Digital Transformational Leadership in maturing Green Digital Mindset among the followers, which has been regarded as an essential precursor for Green Digital Transformation. Besides expanding the body of knowledge by offering rare insights into the interplay among Green Digital Transformational Leadership, Green Digital Mindset, and Green Digital Transformation, the newfangled model provides significant implications for the organizations'', teams', and/or leaders' pursuits of capacitating green digital transformation through the culmination of green digital mindset among the followers, and the cultivation of an enabling green digital culture.

1. Introduction

The green transformation concept, while emphasizing the significance of sustainability and environmental responsibility (Abbas and Khan, 2023; Withisuphakorn et al., 2019), signifies the making of conscious choices by the organizational leadership to truncate the (adverse) effects on the environment, conserve natural resources, and protect the planet for future generations, through encouraging the adoption of sustainable practices while promoting a culture of environmental responsibility (Huang et al., 2023; Lamba et al., 2023). The process entails the establishment of a collective vision and harmonizing efforts to promote ecological sustainability, cooperation, originality, and consistent integration and enhancement of sustainable methodologies, technologies, and practices within an organization (Dhir et al., 2023; Iqbal and Piwowar-Sulej, 2023). As organizations all over the world are progressively seeking to adopt green practices and digital innovation, understanding how leaders inspire and intellectually stimulate their followers/teams toward sustainable digital efforts has become quite crucial and has attracted significant research and managerial attention in recent times (AlNuaimi et al., 2022; Al-Swidi et al., 2021; Talwar et al., 2022).

As a pro-active and essential response to the emerging environmental

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issues, the idea of green digital transformational leadership (hereafter, GDTL) has emerged, which emphasizes the role of leaders in not just prioritizing green initiatives but also enhance followers' technological readiness, spark creativity, and enable teamwork to actualize long-term organizational through an effective orchestration of green digital transformation (Ali et al., 2023; Li et al., 2020). The GDTL approach centers on motivating and allowing employees to adopt ecologically conscious conduct and advocate for sustainability in their professional endeavors (Sidney et al., 2022; Weber et al., 2022). This leadership style's salient characteristics include sustainability, cooperation, ingenuity, and continuous improvement (Althnayan et al., 2022; Awan et al., 2023). The implementation of GDTL entails the creation of a conducive organizational culture, advocating for environmental consciousness and education, and cultivating employee involvement and engagement in sustainability endeavors (Begum et al., 2022b; Riva et al., 2021). The prime focus of this leadership approach is to establish a collective vision for ecological sustainability and inspire employees to undertake measures that could facilitate the realization of this vision (Singh et al., 2020; Zhu et al., 2022).

GDTL requires a potent realization and advocacy about the significance of environmental sustainability amongst all employees (Cop et al., 2020). This includes encouraging environmentally responsible actions and creating avenues for employee involvement and engagement in sustainability initiatives (Rizvi and Garg, 2021; Suliman et al., 2023). GDTL, thus, encompasses various interrelated facets, such as the influence of organizational culture on the formation of employees' attitudes and conduct towards environmental sustainability and the significance of proficient communication and leadership in promoting sustainable transformation (Liu and Song, 2023; Sun et al., 2022). This leadership style has been found to profoundly enhance employee engagement and motivation to embrace environmentally responsible behaviors (Bhatt et al., 2020; Iqbal and Piwowar-Sulej, 2023).

According to a recent report by Accenture.com (Accenture, 2022), integrating digital technologies can accentuate digital transformation and promote environmental sustainability. The report suggests that digital technologies can potentially decrease global carbon emissions by as much as 20% (Accenture, 2022). A survey conducted by Deloitte.com (Deloitte, 2022) found that a significant proportion of millennials (75%) consider a company's environmental and social commitments when deciding about their employment, indicating a growing inclination towards preferences for sustainability in the professional sphere. Furthermore, Nicoletti Jr. et al. (2022) contend that organizations that demonstrate a robust dedication to sustainability exhibit superior performance compared to their counterparts over an extended period. Research has also shown the efficacy of GDTL in fostering sustainable practices within organizations (Chen and Yan, 2022; Malodia et al., 2023). (Begum et al. (2022b), and Farrukh et al. (2022) reveal that the implementation of GDTL has the potential to induce a change in the cognitive orientation of employees, encouraging them to adopt digital technologies that support ecological sustainability (Chu et al., 2023; Farooq et al., 2021). This, in turn, can foster sustainable digital transformation within organizational settings (Bhutto et al., 2021; Chaudhary et al., 2023). Due to its momentous efficacy to simultaneously affect desirable outcomes in the digital and ecological transformation domains, GDTL has gained significant attention from both researchers and practitioners in recent times (Chaudhary et al., 2023; Muisyo et al., 2023).

It has been proposed that GDTL can be implemented effectively in different organizational contexts (Singh et al., 2020) and can lead to measurable improvements by increasing focus on green digital transformation (GDT) (Kristoffersen et al., 2021; Pan and Nishant, 2023) by creating an organizational green digital culture (Rizvi and Garg, 2021; Yeşiltaş et al., 2022) and employees' green digital mindset (Hooi et al., 2022; Raza and Khan, 2022). Employee green digital mindset (hereafter, GDM) pertains to employees' cognitive dispositions, convictions, and principles in the context of utilizing digital innovations to advance ecological sustainability (Iqbal and Piwowar-Sulej, 2023; Leso et al., 2023). The GDM of employees encompasses the promotion of digital literacy and the cultivation of awareness regarding the environmental ramifications of digital technologies among the workforce (Ly, 2023; Singh et al., 2020). Green digital transformation (hereafter, GDT) envisages digital technologies to facilitate ecological sustainability and advance sustainable practices within organizational settings (Weber et al., 2022). This entails utilizing technological advancements, such as automation, data analytics, artificial intelligence, etc., to mitigate environmental effects, enhance resource utilization, and foster sustainable practices throughout an enterprise's activities (Borowski, 2021; Ruan et al., 2022). Thus, GDT commends the integration of sustainable digital practices enabled by modern digital technologies and the advocacy of responsible leadership practices (Feng et al., 2022). The prime objective of GDT is to establish a sustainable and ecologically accountable organization through an adequate integration of digital technologies to facilitate sustainable actions at the micro, meso, and macro levels (Feroz et al., 2021).

The convergence of leadership (style), (employee) attitude, and sustainability (outcomes) have gained profound centrality in the scholarly discourse in this digitalization and sustainability-driven age. Using the Stimuli-Organism-Response (SOR) framework and the transformational leadership model as the underpinning theories, this study seeks to contribute to the burgeoning scholarly discourse by examining the interactions between GDTL and GDM to discern how they affect the course of green digital transformation (GDT) under varying permutations of organizational green digital culture (hereafter, OGDC). Focusing on the need for further research to discover the factors affecting the acceptance and execution of green practices by organizations, particularly in developing countries (Widisatria and Nawangsari, 2021; Zhao and Huang, 2022), the pivot of this study is the mediating role of GDM of employees in the relationship between GDTL and GDT. Besides, it seeks to unravel the fundamental mechanisms and boundary conditions by examining the moderation effect of OGDC (Luu, 2022; Özgül and Zehir, 2022; Şengüllendi et al., 2023) in causing GDTL to affect GDM (Alrasheedi et al., 2022; Gilch and Sieweke, 2021). As such, the study endeavors to answer the following research questions.

RQ1: Does GDTL indeed enhance employee GDM?
RQ2: How can GDTLs impact the successful orchestration of GDT?
RQ3: Does GDM mediate the impact of GDTL on GDT?
RQ4: How does OGDC augment the development of GDM in employees through GDTL?

Through providing answers to these research questions, the contribution of the current study is twofold: 1) This research integrates and extends the transformational leadership theory (Bass and Avolio, 1993) with stimuli-organism-response model (Mehrabian and Russell, 1974) to the sustainably context to fathom the efficacy of GDTL in accentuating GDT through enhancing employee GDM under disrate contextual conditions. 2) This study makes vital theoretical and practical contributions by elucidating upon the mechanism and contextual conditions girdling employees' perceptions and receptivity to their GDT leaders' efforts in combating challenges inherent in orchestrating green digital transformations in their organizations (Begum et al., 2022a; Farrukh et al., 2022). Additionally, the research could provide useful insights into employees' attitudes toward digital and green transformation and the role of GDTL in effectuating these attitudes (Brunetti et al., 2020; Olya et al., 2019). Another contribution of this study could be traced back to its usage of Middle Eastern data, which could not only provide rare insights into the dynamics of green digital transformation in this unique Saudi Arabian context which is currently undergoing a social, digital, and ecological transformation under its overarching Vision 2030. Badghish et al. (2023), but could also cement generalizability of the etic theories coined in the West to other socio-cultural settings.

The remainder of the paper is structured like so. Section two presents a review of the extant literature along with the conceptual model and the study hypotheses. Section three discusses methodology. Section four rolls out the results. Section five organizes a discussion stemming from these results and highlights this study's theoretical and practical contributions. The final section concludes the entire discussion, duly highlighting its limitations and suggesting some avenues for future research.

2. Literature review

A burgeoning corpus of scholarly works has posited that the concepts of green and environmental sustainability are gaining paramount prominence in organizational decision-making (Çop et al., 2021; Hussain et al., 2022). Besides, there is increasing scholarly attention on the significance of leadership in facilitating sustainable digital transformations within corporate entities through fostering environmentally conscious attitudes and conduct at all levels (Begum et al., 2022b; Wang et al., 2021). The GDTL approach has quickly emerged as a practical approach towards advancing ecological sustainability and facilitating sustainable transformation (Awan et al., 2023; Sun et al., 2022). Similarly, the implementation of (Industry 5.0) digital technologies can profoundly augment the process of achieving sustainable digital transformation and promoting ecological sustainability (Porfírio et al., 2021; Yaqub and Alsabaan, 2023).

Green and digital activities complement each other in numerous ways, contributing to a more sustainable future (Rosário and Dias, 2022). The convergence of these two has paved the way for innovative solutions and transformative practices that benefit the planet and society (George et al., 2021). One of the most significant contributions of the digital realm to the green movement is the increased accessibility to information (Wang et al., 2021). Internet and digital technologies have revolutionized how we gather and disseminate knowledge about environmental issues (Feroz et al., 2021; Hsu et al., 2020). People can now access vast amounts of information on sustainable practices, renewable energy solutions, conservation efforts, and eco-friendly products with only a few clicks (Valerio-Ureña and Rogers, 2019). This ease of access to information empowers individuals and organizations to make informed choices and take proactive steps toward reducing their ecological footprints (D'Amato et al., 2019; Mehmood et al., 2022). Besides, digital technologies have played a crucial role in optimizing resource consumption and minimizing waste generation (Kurniawan et al., 2022). Additionally, advanced data analytics and machine learning algorithms are employed to optimize industrial processes, transportation networks, and supply chains, significantly reducing energy consumption and greenhouse gas emissions (Zhang et al., 2021).

The digital realm has also provided a platform for raising awareness and mobilizing collective action on environmental issues (Tim et al., 2018). Digital technology platforms have become powerful tools for environmental advocacy, allowing individuals and organizations to amplify their voices and reach a global audience (Zawieska et al., 2022). Green digital technologies, platforms, petitions, and crowdfunding initiatives have driven environmental conservation and green efforts, supported reforestation projects, and promoted sustainable practices (Butticè et al., 2019). The digital sphere has fostered a global community, connecting like-minded individuals and organizations across borders to collaborate and work towards common environmental goals towards a green digital mindset (Liu and Song, 2023). On the other hand, green initiatives have also benefited from digital technologies. The increased adoption of renewable energy sources such as solar and wind power has been made possible through advancements in digital monitoring, control systems, and energy storage solutions (Wang et al., 2021b). These digital advancements enhance the efficiency and effectiveness of green technologies, making them more viable and attractive options for sustainable living and transforming our lives (Zhang et al., 2023).

GDTL, which amalgamates green, digital, and transformational approaches to leadership, signifies a visionary and innovative perspective that facilitates sustainable change within organizations and society through green, digital, and/or technological activities (Sidney et al., 2022; Al-Ghazali et al., 2022). Leaders with a profound comprehension of environmental challenges and the potential of digital technologies to tackle them are highly adept (Gilli et al., 2023; Yang et al., 2022). They motivate and enable their teams to adopt environmentally sustainable practices, leverage digital technologies, and generate beneficial influences on the natural world (Wang et al., 2021). They use progressive methodologies that foster the implementation of environmentally friendly technologies, facilitate cooperation, and propel the shift toward a more ecologically sound future (Stanley et al., 2021). This holds the vision of utilizing digital innovations to address environmental concerns (Zawieska et al., 2022; Feroz et al., 2021). Their leadership style is transformative and serves as a source of inspiration for others (more specifically, the followers) to adopt similar practices, thereby generating a ripple effect of favorable outcomes in the realm of eco-friendly digital pursuits (Zafar et al., 2022; Li et al., 2020b).

The complementarity between green and digital activities is evident in how they reinforce and amplify each other's impact (Yang et al., 2022). Transformational leaders can use digital technologies to provide information, optimize resource consumption, foster collaboration, and raise awareness (Philip, 2021). Green initiatives benefit from digital advancements in improving efficiency, monitoring, and control (Lee et al., 2023). Together, they form a powerful alliance that has the potential to accelerate the transition toward a sustainable and environmentally conscious future (Feroz et al., 2021; Stanley et al., 2021). The contemporary literature unravels that an emerging strategy for improving ecological sustainability and facilitating sustainable transformation in organizations is the green digital transformational leadership approach (Awan et al., 2023; Sun et al., 2022), wherein the use of digital technology by leaders can enhance the process of achieving sustainable digital transformation while encouraging ecological sustainability (AlNuaimi et al., 2022; Liao et al., 2023).

There has been mounting interest in studying GTL and its effect on companies. A snapshot of the trend in previous studies carried out in GTL is presented in Fig. 1, which brings forth the progressive attention being paid by researchers in this area. As can be seen in Fig. 1, research in GTL is a relatively recent phenomenon, with only one paper being published yearly till 2019. The number of papers being published yearly reached a figure of 7 papers and above in the year 2020. Growth in the number of research publications in this area reached double digits in 2021, with the number of papers published becoming more than triple (31) in 2022 compared to the number of publications in 2021 (10). The number of publications has increased to 34 in 2023, showing the growing interest of academia in the field of GTL. Hence, there is growing scope for increased research in this area. However, despite the increasing interest in GTL, there is a scarcity of empirical research examining the effectiveness of this leadership style in promoting environmental sustainability in organizations (Farrukh et al., 2022; Li et al., 2020; Odugbesan et al., 2023).

Despite their commendable contributions in advancing the scholarly discourse, the previous studies related to GDTL feature several gaps. Firstly, the studies are limited to studying the GTL style in specific industrial contexts like manufacturing SMEs (Farrukh et al., 2022; Singh et al., 2020), restaurants (Tosun et al., 2022), grocery, food, personal care (Hameed et al., 2022), hospitality (Khanra et al., 2021; Öğretmenoğlu et al., 2022) and others (Odugbesan et al., 2023). The current study extends the application of GDTL in the multi-industry context. Secondly, no studies have explicitly focused on the Middle Eastern or Saudi Arabian contexts despite the great potential to examine the dynamics of green and digital transformation in these contexts. Our research focuses exclusively on examining the impact of the GDTL style in Saudi Arabian organizations. Thirdly, the (mediating or moderating) conditions that have mainly been employed while studying the role of GDTL have mostly been related to green human resource management practices (Farrukh et al., 2022), green performance (Tosun et al., 2022),

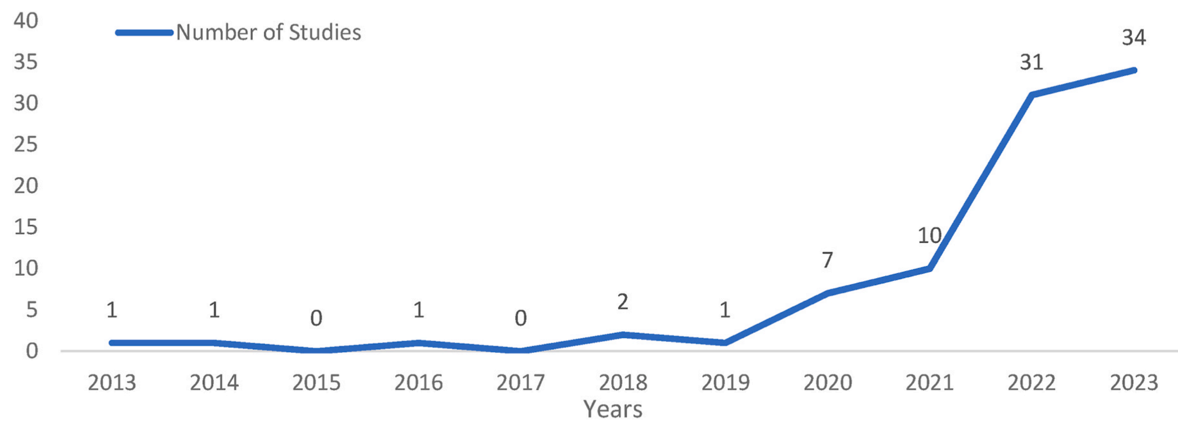

Fig. 1. Number of Studies on GTL over the years (Source: Authors' work).

and perceived organizational support (Hameed et al., 2022), among others. Consequently, our use of GDM and OGDC as mediating and moderating conditions in examining GDTL-GDT linkage seems quite new-fangled (Abbas and Khan, 2023; Azhar and Yang, 2022). The underpinning theories for our conceptualization of the relevant associations among the subject construct in our study have been the Stimulus-Organism-Response (hereafter, SOR) framework and the transformational leadership theory.

2.1. The underpinning theories

2.1.1. The Stimulus-Organism-Response (SOR) framework

The SOR framework suggests that external stimuli affect individuals, subsequently shaping their behaviors (Mehrabian and Russell, 1974). The framework postulates that stimuli like GDTL style (Dhir et al., 2021; Lee et al., 2023; Timpanaro, 2023) and OGDC (Chou et al., 2022) may influence employees' (organism) perceptions and attitudes (including GDM), eventually shaping their behavioral responses (Saleem et al., 2020; Sun et al., 2022), which would be their participation in GDT initiatives. The strategy of digital green transformation seeks to advance sustainable environmental practices through digital technologies (Carayannis and Morawska-Jancelewicz, 2022; Paiola et al., 2021; Pathak et al., 2024). In alignment with the SOR framework, adopting and implementing green digital transformation driven by modern technologies acts as an external stimulus (Gupta and Bose, 2022; Priyono et al., 2020). This influences employees' internal organizations, such as their attitudes and perceptions about digital green transformation (Du and Yan, 2022; Liu and Song, 2023), impacting their readiness to adopt digital technologies for environmental sustainability (Qu et al., 2022; Widisatria and Nawangsari, 2021). The SOR framework forms the theoretical foundation for understanding the relationships between GDTL, GDT, and employees' behavioral responses via their green digital mindset. Using this framework, our study aims to decipher how GDTL and GDM influence GDT and foster ecologically responsible actions at work.

2.1.2. Transformational leadership theory

TL theory (Bass and Avolio, 1993) signifies transformational leadership as a management concept that inspires employees to devise new approaches to expand and enhance an organization's success (Afsar and Umrani, 2020). It seeks to create a vision, effectively communicate it to followers, and then align its followers' goals and values with that vision (Steinmann et al., 2018). The critical facets of such a leadership style include idealized influence, inspirational motivation, individualized consideration, and intellectual stimulation (Bass, 1985; Zafar et al., 2023). TLs can inspire and empower followers by enabling the work environment for growth, development, and success (Nuel et al., 2021). Transformational leaders initiate cultural change by understanding the existing culture, reorienting it toward a novel vision, and reevaluating collective assumptions, values, and norms (Bass and Avolio, 1993; Özgül and Zehir, 2022). Integrating transformational leadership with green and digital leadership results in GDTL, which emphasizes motivating employees to engage in behaviors that contribute to long-term environmental viability (Chams and García-Blandón, 2019; Widisatria and Nawangsari, 2021).

OGDC is the fundamental aspect that establishes the core values, beliefs, and standards that shape an organization's approach to green digital transformation (Trapp and Kanbach, 2021; Pan et al., 2022). It cultivates a mentality that gives importance to sustainability and environmental awareness, molding the decision-making process and impacting the company's behavior (Zhang et al., 2023; Al-Ghazali et al., 2022). While in the other hand, GDM cultivated within the framework of GDC enables individuals to actively participate in green digital transformation endeavors (Martínez-Peláez et al., 2023; Liao et al., 2023). It fosters a conviction in the capacity of digital technologies to tackle environmental issues and motivates individuals to pursue creative solutions (Liu et al., 2022; Samuel et al., 2022). Besides that, GDT, fueled by GDC and GDM, manifests as a tangible shift in operational practices and technological adoption (Aboelmaged and Hashem, 2019; Al-Swidi et al., 2021). It encompasses implementing environmentally friendly digital solutions, promoting sustainable resource management, and reducing environmental impact (Aftab et al., 2022; Althnayan et al., 2022). These three components are inextricably linked, forming a synergistic loop that drives sustainable progress. GDC shapes GDM, which in turn informs GDT. As GDT unfolds, it reinforces GDC and GDM, creating a continuous and sustainable improvement cycle.

To more effectively implement GDT, exploring its multifaceted aspects and implications is crucial (Vial, 2019; Xue et al., 2022). Research in this field can shed light on the strategies, challenges, and outcomes associated with GDT initiatives (Brunetti et al., 2020; He and Su, 2022). Moreover, examining the role of leadership, organizational culture, stakeholder engagement, and technological advancements in facilitating GDT can offer insights for practitioners, academics, and policymakers (Ciasullo and Lim, 2022; Rosário and Dias, 2022). Continuous academic investigation and analysis of GDT can enhance understanding of its significance, consequences, and capacity to foster a sustainable and technologically advanced future (Cheng et al., 2023; Gomez-Trujillo and Gonzalez-Perez, 2021). Such research can aid organizations in navigating this transformative process, making informed decisions that contribute to positive environmental change while harnessing the potential of digital technologies. Thus, our current study adds to previous knowledge and theory on GDT by highlighting the impact of GDTL and GDM on GDT in an organization. Additionally, the role of GDM in the relationship has also been investigated.

Despite the growing interest in GDTL, empirical research examining its impact on GDC creation, GDT facilitation, and its relationship with

GDM is limited. This study addresses this gap by investigating these relationships, specifically exploring how GDTL impacts GDM development and GDT adoption (Fig. 2). Furthermore, the study explores the moderating role of GDC in the relationship between green digital transformational leadership and the mediating role of employees' GDM.

The following section discusses the nature, scope, and hypothesized associations among our conceptual model's subject constructs.

2.2. The study constructs

2.2.1. Green digital transformation (GDT) - the outcome

Green Digital Transformation (GDT) refers to integrating digital technologies into an organization to enhance environmental sustainability (Liao et al., 2023) by encouraging eco-friendly innovation, maximizing resource consumption, and reducing waste (Feng et al., 2022; Weber et al., 2022). The activities under it include integrating intelligent building technology, adopting energy-efficient equipment, collaborating virtually to cut down on travel, and using data analytics to optimize resource use, minimize waste, and promote eco-friendly innovation (Feng et al., 2022; Zoppelletto et al., 2023). GDT signifies a fundamental shift in business models, aiming to align strategies with environmental sustainability objectives (Feroz et al., 2021; Leso et al., 2023). Modern organizations recognize the importance of sustainable practices in the evolving digital landscape (Eisner et al., 2022; Mousa and Othman, 2020). GDT emerges as a transformative strategy that blends digital tech and sustainability to achieve ecological goals (Gomez-Trujillo and Gonzalez-Perez, 2021; Robertsone and Lapiņa, 2023). Organizations can leverage digital tools to enhance operations, reduce environmental impact, and move towards a more sustainable future (Ruan et al., 2022; Xiong et al., 2022).

2.2.2. Green digital transformational leadership (GDTL)- the antecedent

GDTL is a leadership approach that intends to create a sustainable and environmentally conscious organizational culture (Farrukh et al., 2022; Weber et al., 2022). The GDTL approach goes beyond standard forms of leadership, which only emphasizes the importance of attaining organizational goals. Instead, it places an emphasis on the significance of being a good steward of the environment (Khan and Khan, 2022; Zhao and Huang, 2022). This is accomplished by motivating and empowering employees to develop environmentally friendly practices and behaviors (Ly, 2023). This style of leadership encourages the preservation of the natural environment by motivating subordinates to alter the attitudes, values, and actions they have towards it with the goal of creating a greener and more sustainable organization (Rizvi and Garg, 2021), Providing employees with the support and resources they need to adopt environmentally friendly practices, cultivating a culture of environmental responsibility within the organization, and establishing a clear vision and goals for environmental sustainability are all aspects of green digital transformational leadership (Ansari et al., 2021; Özgül and Zehir, 2022). This, in turn, results in advantages for both organizations and society (Tosun et al., 2022).

2.2.3. Employees' green digital mindset (EGDM)

EGDM pertains to individuals' consciousness, comprehension, and motivation to utilize digital transformation ecologically sustainably (Chen and Yan, 2022). The adoption of environmentally responsible habits when using digital technology and the support of ecologically sustainable practices are integral components of GDM (Ansari et al., 2021; Pradana et al., 2022). Culminating EGDM may be an important precursor to GDT as it could catalyze employees' participation in GDT initiatives.

2.2.4. Organizational green digital culture (OGDC)- the moderator

OGDC encompasses shared values, beliefs, and behaviors prioritizing sustainability through digital technologies (Rizvi and Garg, 2021; Zhen et al., 2021). It fosters environmentally conscious practices using digital tools, encouraging employees' participation in sustainability initiatives (Kim et al., 2020; Weber et al., 2022). GDM-related practices include deploying green technology, developing sustainability policy using digital platforms, and involving staff in digital sustainability projects (Pradana et al., 2022; Wang et al., 2021).

2.2.5. The control variables

Wang et al. (2023) stated that organization size, age, staff age, industry nature, qualification, and background influence environmental activities. Thus, this research used organization size (OS) and participants' age (age) as control variables.

2.3. Hypotheses of study

2.3.1. GDTL, GDM and GDT

The synergy of green and digital activities can foster a more sustainable future (Rosário and Dias, 2022) by driving innovative practices benefiting the planet and society (George et al., 2021). With the internet and technology revolutionizing knowledge sharing on environmental concerns (Feroz et al., 2021; Hsu et al., 2020), digital advancements have considerably improved accessibility to environmental information (Wang et al., 2021). People may easily access a wealth of information and knowledge about sustainability, renewable energy, conservation, and eco-friendly items with only a few clicks nowadays (Valerio-Ureña and Rogers, 2019), enabling them to make well-informed decisions and engage in environmentally favorable actions (D'Amato et al., 2019;

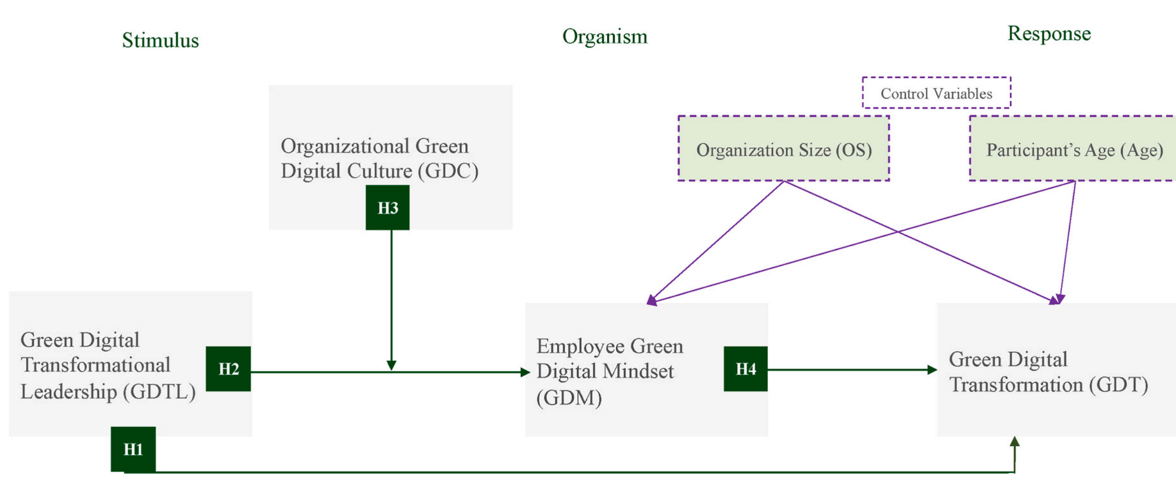

Fig. 2. The conceptual model.

Mehmood et al., 2022). As a result, digital technology can support organizational green activities and encourage using renewable energy sources (Wang et al., 2021). However, for effective synergy of green and digital activities in organizations, leadership is crucial (Ly, 2023; Zhao and Huang, 2022). Several recent studies, such as Farrukh et al. (2022), and Odugbesan et al. (2023) have highlighted the efficacy of green leadership and/or digital leadership in producing desirable performance outcomes in various contexts. Besides, Zhao and Huang, 2022) have profoundly elucidated the instrumentality of effective leadership in materializing digital and sustainability outcomes through creating a synergy of green and digital activities in the organizations. Extending insights from these studies to the GDT context, we hypothesize.

H1. GDTL has a direct positive association with (successful implementation of) GDT.

Though previous studies have studied the association of GDTL and certain behavioral outcomes in several contexts (Farrukh et al., 2022; Odugbesan et al., 2023; Singh et al., 2020), there has been an acute shortage of research analyzing the role of GDTL in maturing EGDM and its consequent impact on green digital transformation (GDT) of an organization. An organization's sustainability journey can be seen holistically by examining the relationship between GDTL, employee GDM, and organizational GDT, which can also offer important insights into the tactics, attitudes, and values that support a thriving, environmentally conscious digital transformation. Ly (2023) highlights the role of leadership in motivating and empowering employees to develop environmentally friendly practices and behaviors. Rizvi and Garg (2021) reveal green digital leadership encourages preserving the natural environment by motivating subordinates to alter their attitudes, values, and actions towards it to create a greener and more sustainable organization through effective integration and exploitation of digital technologies. Besides, researchers like Ansari et al. (2021) and Özgül and Zehir (2022) contend that by providing employees with the support and resources they need to adopt environmentally friendly practices, GDTLs could cement their orientation, passion, and commitment for green digital initiatives. In accordance with these assertions, we hypothesize.

H2. GDTL has a significant positive association with employee GDM.

2.4. EGDM and GDT

An individual with a green digital mindset (GDM) uses digital technology with awareness and consideration for the environment (Ojo et al., 2019; Uvarova et al., 2021). It entails employing digital technology to assist sustainability while implementing eco-friendly practices (Ansari et al., 2021; Pradana et al., 2022), highlighting the contribution of digital technologies to sustainability and value (D'Amato et al., 2019; Singh et al., 2020). This is accomplished through more eco-aware company practices in the digital age, less energy use, and decreased carbon emissions (Zhen et al., 2021). Technology is critical in translating these values, significantly impacting people's willingness to employ in green activities (Hooi et al., 2022; Lusiani et al., 2020). Developing GDM involves adopting eco-responsible digital behaviors, considering the environmental impact of technology, and advocating for sustainability in digital business (Feroz et al., 2021; Samuel et al., 2022). By reducing energy waste, this mentality helps the environment by reducing the carbon footprint of the digital industry (Chen and Yan, 2022; Katz et al., 2022). Employees are influenced by green and digital transformational leadership (GDTL), preparing them for change (Özgül and Zehir, 2022; Ruan et al., 2022). The organization's growth is fueled by this interaction, which results in a seamless alignment of digital environmental goals with organizational goals (Dittes et al., 2019; Du and Yan, 2022). The current study examines the mediating role of GDM in the interaction between visionary leaders and organizational journeys to discover the connection between GDTL and GDM further. Therefore, the following two hypotheses were developed.

H3. Employees' GDM has a significant positive association with GDT.

H4. Employees' GDM mediates the relationship between GDTL and GDT.

2.5. The moderating role of OGDC

Through digital technology, OGDC produces advantages like improved participation, reputation, stakeholder relationships, cost savings, efficiency, and reduced environmental impact (Bass and Avolio, 1993; Qu et al., 2022). Previous studies have explored the influence of GDC on the association between green core competencies, green absorptive capacity, and digital tech and have highlighted that nurturing green core competencies is crucial for green innovation, with a supportive GDC playing a pivotal role in fostering green absorptive capacity through digital tools (Abbas and Khan, 2023; Isensee et al., 2020).

A supportive OGDC plays a crucial role in fostering green absorptive capacity through digital tools and is essential for fostering sustainability in businesses. It is entwined with digital culture (Roscoe et al., 2019) with green digital transformational leadership encouraging and empowering employees to adopt this green culture. However, the link between the GDC and GDTL is intricate and extends beyond simple effects (Zhang et al., 2021; Zhen et al., 2021). GDC can operate as a mediator intimately linked to GDTL to affect the creation and manifestation of a Green Digital Mindset (GDM) in the workforce (Hu et al., 2023; Sidney et al., 2022).

This interactive process ingrains sustainability principles within the organization wherein aligned values foster a conducive environment for GDM development (Qu et al., 2022; Zhen et al., 2021), driving ecologically sound decision-making and green digital transformation via digital technologies. Despite previous studies, the intricate association between GDC, GDTL, and GDM calls for further empirical and theoretical investigation. Our research aims to analyze the mechanisms through which organizational context and leadership jointly drive sustainable practices, digital transformation, and green initiatives using digital technologies. Hence, the fifth hypothesis is proposed as follows.

H5. OGDC moderates the relationship between GDTL and employees' GDM.

3. Research methodology

3.1. The measurement scales

The measurement instrument (the questionnaire) consisted of two parts. The first part contained the measurement scales for the four constructs adapted from previous studies (Appendix Table A1). GDTL was measured using six items adopted from (Chen and Chang, 2013). Six items to measure OGDC were adopted from Marshall et al. (2015). For GDM, seven items were adapted from Steg et al. (2005). Finally, the GDT was measured using a three-item scale adapted from Aral and Weill (2007). All responses were recorded using a seven-point Likert scale (Fornell and Larcker, 1981), with one point denoting "strongly disagree" and seven points denoting "strongly agree." The second section of the questionnaire consisted of measures to capture the participants' demographical information, including their profession, occupation level, department, qualification, company size, industry, location, age, gender, and work experience (Appendix Table A2). A couple of experts from the sector conducted a review of the questionnaire before its deployment. The questionnaire was translated into Arabic with subsequent back-translation to English, as recommended by Brislin (1980), to ensure consistency in their clarity and meaning.

3.1.1. The pilot testing

Before administering the study to the intended participants, to ensure the validity and applicability of the questionnaire, a pilot study was steered with a select group of individuals as recommended by (Boote, 1981; Cronbach, 1971; Reynolds and Diamantopoulos, 1998). The pilot study contained 40 samples for "Cronbach's alpha (CA)" testing, and all elements' values met the threshold of more than 0.7. The "coefficient of determination (R^2)" in the pilot study was substantial for both GDM and GDT (Hair et al., 2021). Thus, all items were retained in the questionnaire for formal data collection.

3.2. Sampling and data collection

The sampled population for this study is executives working in various firms in the Saudi Arabian industry. Using the a priori method, we determined the sample size for the structural model (Soper, 2023). The minimum required sample size to detect the desired effect was determined to be 137, given the medium effect size of 0.3 (Cohen, 1992), with a targeted statistical power level of 0.80, using four latent variables measured through 22 items at a significance level of 0.01. Thus, the final sample size (n = 240) was larger than the suggested minimum of 137. A significantly larger sample size was selected to decrease the margin of error and bias and obtain a higher level of accuracy with higher confidence and insightful information (Gumpili and Das, 2022).

A questionnaire-based survey method was adopted for the collection of data. The participants for the study were selected and approached via LinkedIn using the convenience and snowball sampling method known for its efficiency in obtaining timely responses (Al-Swidi et al., 2021). Data was collected using web-based questionnaires and a note on the research objective to provide a basic idea about the proposed research. The questionnaires were distributed through one of the author's LinkedIn profiles, employing the messaging feature of the site. Five hundred messages were sent to targeted participants via the LinkedIn message feature.

The LinkedIn profile of one of the authors with 21 years of professional experience, with around 30,000 network members, made it possible to reach the desired sample for data collection. A message was sent to the relevant connections using their first names to emphasize the electronic questionnaire's importance, increase the likelihood of participation, and facilitate completion. Using web-based questionnaires was employed because it is a technology that streamlines and speeds up the process, saving money, time, and effort. (Fowler, 2013; Weber et al., 2005).

LinkedIn is considered a powerful tool for reaching targeted samples and effective data collection in business research (Alabdali and Salam, 2022). The initial phase involved the identification of the intended recipients and the development of communication that effectively elucidated the objectives and advantages of the investigation. Leveraging the messaging functionality of LinkedIn as a means of gathering data can yield favorable outcomes in terms of engaging with a sizable pool of prospective respondents (Cerruto et al., 2022; Ebert et al., 2018; Mahmoud et al., 2022). To optimize response rates, follow-up messages were also sent to the participants.

3.3. Analytical strategy

The measurement and structural model have been analyzed using a PLS-based structural equation modeling (SEM) technique using SmartPLS 4.0 (Ringle et al., 2022). The reliability and (convergent and discriminant) validity of the measurement model, along with the relevant statistics to assess the hypothesized main, mediating, and moderating effects of the structural model, were obtained/assessed by framing PLS Algorithms and Bootstrapping following Hair et al. (2021). Additionally, PLS blindfolding calculations were employed for the predictive relevance of the test. Following Anderson and Gerbing's (1988) and Ali et al.'s (2023), the analyses were performed in three steps. Performing descriptive statistics was the first step, then computing the measurement model to adjudge validity and reliability, and finally, testing the structural model to see if it could be considered appropriate to test the hypotheses. The following sections present the extensive series of analyses carried out in the present study and the emergent findings.

4. Results

4.1. Descriptive analysis

The response to the demographic variables aids in identifying the profile of the sample of our study, as depicted in Table 1. Women constituted 35% of the sample size, half of the participants were in their senior career level (51%), 70% of the participants were less than 40 years old, 95% of respondents were at a supervisor cadre or above, and 93% had a bachelor's degree or above. These demographic details validate the profiles of the participants as qualified to participate and respond in terms of their experience and maturity to the core elements to be analyzed in the study. Furthermore, one of the features of PLS-SEM is that it can deal with the data without assuming normality (Hair et al., 2021).

4.2. Common method bias/variance (CMB/CMV)

To identify problems with "common method variance" (CMV) in PLS-SEM models, the MLMV methodology ("measured latent marker variable") was used (Chang et al., 2020). Years of experience was a measured variable that did not belong to the same domain as the other variables; thus, it was included in the model. Table 2 shows that the R^2 values of both GDM and GDT did not differ after adding the years of experience (YoE) (Fig. 3), which suggested that the model of this study did not have CMB/CMV issues (Terho et al., 2022; Yoshikuni and Albertin, 2020). Additionally, a random variable was also included in the data set, and the VIF was checked to determine if it was still less than 3.3 for both the inner and outer models (Table 3), which was also confirmed hence no CMV issue as recommendations of Kock (2020).

Table 1
Profile of the respondents.

Respondents (n = 240)	#	%		#	%
Area			*Functions*		
Saudi Arabia	229	96%	Sales	12	5%
Other	11	4%	Marketing	2	1%
Gender			Supply chain	10	4%
Male	155	65%	HR	75	31%
Female	85	35%	IT	23	10%
Age (years)			Strategy	15	6%
20–30	44	18%	Finance	12	5%
31–40	125	52%	Admin	43	18%
41–50	60	25%	Operations	47	20%
51–60	11	5%	*Industry*		
Experience (years)			FMCG	4	2%
<1	2	1%	Healthcare	50	21%
2–5	35	15%	Food & beverage	14	6%
6–10	60	25%	Manufacturing	16	7%
11–15	56	23%	Oil and gas	18	8%
16–20	58	24%	Retails	3	1%
21+	29	12%	IT	7	3%
Education			Telecom	23	10%
High school or less	2	1%	Public sector	9	4%
Diploma	15	6%	Aviation	25	10%
Bachelor	128	53%	Education	5	2%
Master	82	34%	Real estate	5	2%
Ph.D.	13	6%	Banking	9	4%
Occupational Level			Insurance	15	6%
Entry	12	5%	Others	37	15%
Specialist/Supervisor	105	44%	*Organization Size*		
Manager/Sr Manager	77	32%	<100	14	6%
Director	34	14%	100–500	38	16%
Leadership	12	5%	501–1000	23	10%
			1001–5000	69	29%
			>5000+	96	40%

Table 2
Common method bias/variance (CMB/CMV).

	Q^2	R^2 Without Marker Variable	R^2 with Marker Variable
GDM	0.187	0.228	0.237
GDT	0.128	0.217	0.228

4.3. Assessment of the measurement model

4.3.1. Reliability and convergent validity

Items' reliability has been assessed through the significance and values of the factor loadings. All factor loadings were significant and above 0.579, indicating sufficient items' reliability (Hair et al., 2021). The (internal consistency) reliability at the construct levels has been assessed through Cronbach Alpha (CA) and composite reliability (C.R.). All four constructs exhibited adequate reliability since the values of both C.A. and C.R. have been sufficiently above 0.7 (Hair Jr. et al., 2020; Hair et al., 2021). Moreover, Convergent validity has been assessed through AVE, where all the values exceeded the threshold value of 0.50, reflecting sufficient convergent validity (Hair et al., 2021; Sarstedt et al., 2022). Relevant statistics are contained in Table 3.

4.3.2. Discriminant validity

Discriminant validity has been ascertained through Fornell Larcker Criterion (1981) and the heterotrait-monotrait ratio (HTMT) values. Sufficient discriminant validity could be witnessed in Table 4, as the square root of AVE in all instances has been greater than the correlations of the latent variables, as suggested by Fornell & Larcker (1981). Besides, all HTMT values have been less than 0.9 (Franke and Sarstedt, 2019; Rasoolimanesh, 2022).

4.3.3. Assessment of multicollinearity

The "variance inflation factor" (VIF) has been used to evaluate potential collinearity using benchmark VIF <3 (Hair et al., 2021; Kim, 2019; Shrestha, 2020). Table 3 demonstrates that the VIF is less than three for individual constructs; hence, multicollinearity is not an issue.

4.4. Assessment of the structural model

Following recommendations from Hair et al. (2019), the path coefficient, bias-corrected confidence interval, and model-fit measures using bootstrapping 10,000 were obtained to assess the structural model.

4.4.1. Model's goodness-of-fit, explanatory and predictive power

The "standardized root mean square residuals" (SRMR) are reported with a threshold value of less than 0.08, which has been used to assess the model's goodness of fit, as suggested by Henseler et al. (2015).

Table 3
Reliability and convergent validity.

Construct	Item	F.L.	VIF	CA	CR	AVE
Green digital Mindset (GDM)				0.887	0.911	0.593
	GDM1	0.250	1.626			
	GDM2	0.211	2.167			
	GDM3	0.144	1.842			
	GDM4	0.202	1.999			
	GDM5	0.176	2.182			
	GDM6	0.163	2.228			
	GDM7	0.152	2.088			
Green Digital Transformation (GDT)				0.853	0.910	0.772
	GDT1	0.399	1.902			
	GDT2	0.353	2.366			
	GDT3	0.386	2.176			
Green Digital Transformational Leadership (GDTL)				0.882	0.911	0.630
	GDTL1	0.182	1.770			
	GDTL2	0.213	1.671			
	GDTL3	0.218	1.993			
	GDTL4	0.205	2.149			
	GDTL5	0.221	2.577			
	GDTL6	0.220	2.482			
Organizational Green Digital Culture (OGDC)				0.882	0.910	0.629
	OGDC1	0.241	2.086			
	OGDC2	0.252	2.348			
	OGDC3	0.137	1.607			
	OGDC4	0.214	2.183			
	OGDC5	0.251	2.768			
	OGDC6	0.148	1.845			

F.L. Factor Loadings; VIF, Variance Inflation Factor; CA, Cronbach Alpha; C.R., Composite Reliability; AVE; Average Variance Extracted

Table 4
The discriminant validity.

	GDM	GDT	GDTL	OGDC
GDM	**0.770**	0.485[h]	0.425 [h]	0.376 [h]
GDT	0.433	**0.879**	0.381[h]	0.500 [h]
GDTL	0.400	0.331	**0.794**	0.710 [h]
OGDC	0.361	0.447	0.625	**0.793**

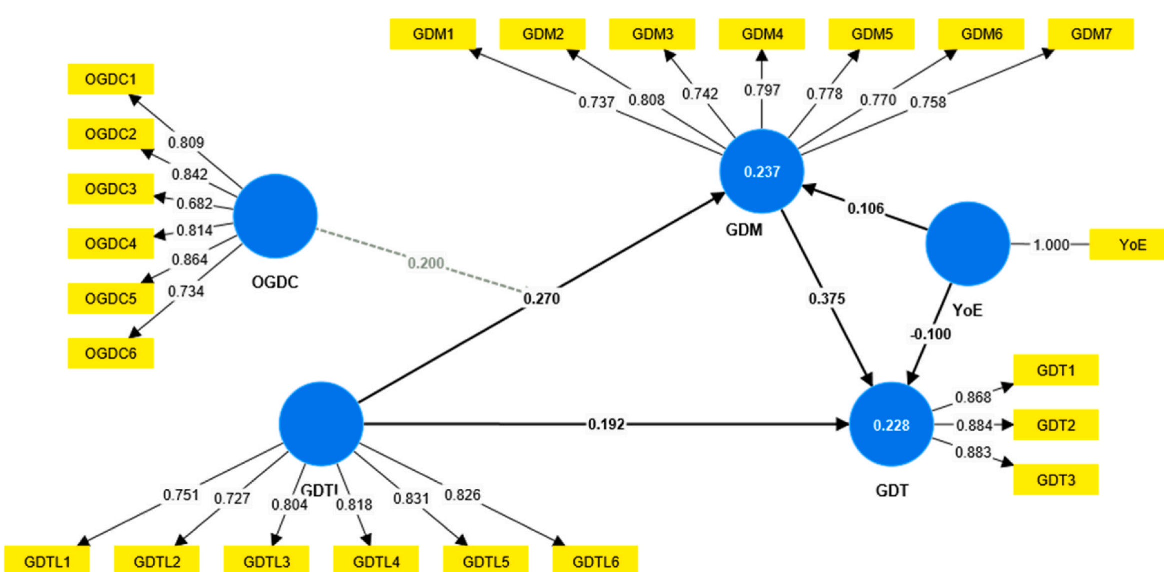

Fig. 3. Marker variable.

Moreover, the squared Euclidean distance (d_{ULS}) and geodesic distance (d_G) were both found to be lower than the upper bound of the CI 95% and 99% (Henseler et al., 2015). The NFI also indicates a good fit for this study's model, as it is close to 1 (Lohmöller, 2013) confirming the model fit. Hence, the model's goodness of fit has been ascertained through all these indices. Relevant statistics are reported in Table 5.

R^2 represents the variation in the endogenous construct explained collectively by all exogenous constructs, which encapsulates the model's ability to explain the variance (Shmueli and Koppius, 2011). The R^2 values of 0.75, 0.50, and 0.25 could be regarded as substantial, moderate, and weak, respectively (Henseler et al., 2009). In the current study, the R^2 value for the endogenous construct was found to be 0.217. This value indicates that approximately 22% of the variance in the endogenous construct is explained by the variance of the exogenous constructs. This result suggests that the model has weak illustrative strength (Chin, 1998; Hair et al., 2019), which is still quite acceptable given the low number of predictors (only two) and the complex nature and context of this study (Zhang, 2022).

f^2 represents the significance of the effect size of each predictor. The threshold values for f^2 of 0.02, 0.15, and 0.35 represent small, medium, and large effects, respectively (Chin, 1998). The f^2 values are 0.038, 0.060, and 0.137, with the effect being between small to medium for H1, H2 and H3. However, all of the results are inside the small, medium, and large threshold ranges of 0.02, 0.15, and 0.35, respectively, as defined by (Cohen, 2013).

Additionally, Chin (2010) suggested that a Q^2 value larger than zero implies that the model is predictively relevant. The PLS Blindfolding calculator was run to determine the predictive relevance of the study. Q^2 values were 0.187 and 0.128 for GDM and GDT, respectively, above 0, indicating that the model's predictive power was well established (Edeh et al., 2023; Wang and Rhemtulla, 2021).

4.4.2. Analysis of the direct, mediating and moderating effects

The path coefficient (β) values were examined to test the impact of exogenous constructs on the mediating and endogenous constructs (see Fig. 4). According to the values presented in Table 6, all three direct effect hypotheses were empirically substantiated. H1 results demonstrate a significant positive relationship between GDTL and GDT ($\beta = 0.188$, $t = 3.107$, and $p = 0.002$. H2 results highlight the significant positive relationship between GDTL and GDM ($\beta = 0.276$, $t = 3.612$, and $p = 0.000$). Similarly, Finally, H3 results corroborate a significant positive relationship between GDM and GDT ($\beta = 0.358$, $t = 5.560$, and $p = 0.000$).

The results of H4 corroborated a significant indirect relationship between GDTL and GDT mediated by GDM ($\beta = 0.099$, $t = 3.748$, and $p = 0.000$). Therefore, the results support the mediation hypothesis, indicating that GDM partially mediates the positive influence of GDTL on GDT. Hence, GDTL not only directly influences GDT but also indirectly influences it through the development of GDM in employees.

H 5 results indicate a significant interaction effect between GDC and GDTL in predicting GDM ($\beta = 0.212$, $t = 3.517$, and $p = 0.000$). Therefore, the results support the hypothesis that the relationship between GDTL and GDM is moderated by GDC, indicating that the effect of GDTL on GDM varies depending on the level of GDC. Fig. 5 encapsulates the essence of this moderation effect.

Table 7 displays the influence of including the control variables on the relationships between the variables. Several noteworthy findings emerged when analyzing the relationships while controlling for other factors. GDM significantly predicted GDT ($\beta = 0.351$, $t = 5.299$, $p = 0.000$). This positive and statistically significant relationship persisted even after considering other variables. In addition, GDTL was significantly associated with GDM and GDT. Controlling for other variables, GDTL predicted GDM ($\beta = 0.291$, $t = 3.761$, $p = 0.000$) and GDT ($\beta = 0.206$, $t = 3.268$, $p = 0.001$).

Furthermore, OGC demonstrated a positive and statistically significant relationship with GDM ($\beta = 0.231$, $t = 2.857$, $P = 0.004$). This association remained significant when other factors were considered. Moreover, the interaction between OGDC and GDTL also exhibited a statistically significant relationship with GDM ($\beta = 0.193$, $t = 3.284$, $p = 0.001$) after controlling for other variables. However, the control variables OS and Age did not show statistically significant relationships with GDM or GDT in either the presence or absence of control variables. In contrast, the control variables OS and Age did not exhibit significant relationships with GDM or GDT, suggesting that they may not substantially impact the observed associations (see Fig. 6).

Table 5
Model fit.

	Saturated model	Estimated model
SRMR	0.071	0.079
d_ULS	1.258	1.576
d_G	0.411	0.428
NFI	0.816	0.810

5. Discussion

5.1. Discussion of results

The findings of this study empirically supported hypothesis one, which postulated a direct positive association between GDTL and GDT. It is in line with some of the previous studies, such as Farrukh et al. (2022) Hameed et al. (2022), Singh et al. (2020), Odugbesan et al. (2023) which attested to the instrumentality of green leadership and/or digital leadership in producing desirable performance outcomes. On the top of it, Ly (2023), and Zhao and Huang, 2022) confirmed efficacy of effective leadership in materializing digital and sustainability outcomes through creating synergy of green and digital activities in the organizations.

Hypothesis two, which postulated a significant positive association between GDTL and GDM, is empirically supported and attests to the efficacy of GDTL in nurturing employees' GDM. This finding is consistent with some of the prior studies that have underscored the significance of leadership in promoting eco-friendly attitudes and/or mindsets. For example, Özgül and Zehir (2022) and Samuel et al. (2022) found that GDTL surpasses conventional methods by significantly emphasizing environmental stewardship and fostering a sustainability mindset by utilizing digital technologies. Feroz et al. (2021) and Ruan et al. (2022) also revealed that GDTLs encourage employees to adopt environmentally responsible practices and behaviors by offering support and resources, promoting a culture of sustainability, and establishing specific sustainability objectives while embracing modern digital transformation, encouraging the development of a green mindset amongst employees wherein technologies can be applied to improve resource efficiency and utilization. Zhu et al. (2022) also concluded that GDTL can affect employees' attitudes and actions concerning the ecological consequences of green digital technology.

The findings of this study also lend credence to hypothesis three, which puts forth a positive association between EGDM and GDT, which means that the cultivation of EGDM facilitates the effective orchestration of GDT. This finding is in line with previous research. Studies conducted by Feng et al. (2022), Peng et al. (2021), Wang et al. (2023), and Xue et al. (2022) also found a positive connection between EGDM and GDT which indicate that the manifestation of GDTL traits by leaders can encourage GDM among followers, thereby increasing the probability of successful materialization of GDT within organizations.

Hypothesis four, which postulated a positive mediation of EGDM in causing GDTL to produce desired effects on GDT, has also been corroborated by the findings of this study. This findings is consistent with similar findings of some of the previous studies Chen & Yan (2022), Feroz et al. (2021), Katz et al. (2022), Özgül and Zehir (2022), Rosário and Dias (2022), Ruan et al. (2022), and Samuel et al. (2022) which

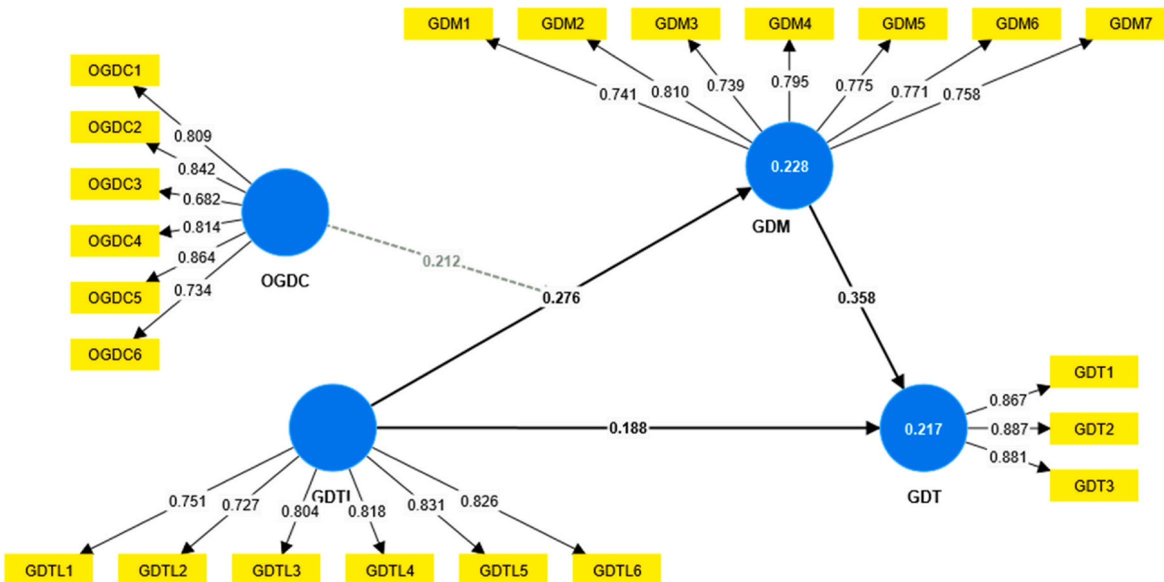

Fig. 4. Structural model result.

Table 6
Assessment of the structural model.

Relationship	β	Std Error	t-values	CI 2.5%	CI 97.5%	p-values	Decision
H1: GDTL - > GDT	0.188	0.061	3.107	0.071	0.311	0.002**	Supported
H2: GDTL - > GDM	0.276	0.076	3.612	0.129	0.429	0.000***	Supported
H3: GDM - > GDT	0.358	0.064	5.560	0.232	0.482	0.000***	Supported
H4: GDTL - > GDM - > GDT (Mediation)	0.099	0.026	3.748	0.050	0.153	0.000***	Supported
H5: OGDC x GDTL - > GDM (Moderation)	0.212	0.060	3.517	0.082	0.318	0.000***	Supported

Notes: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

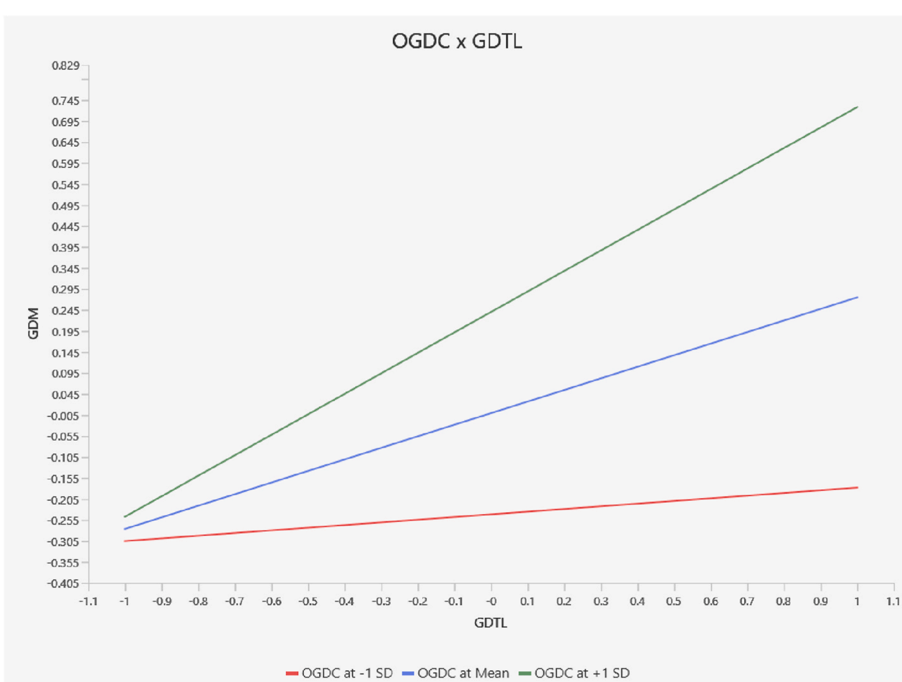

Fig. 5. The slop graph of moderation.

investigated the instrumentality of employee attitudes in enabling leaderships' efforts in producing desirable sustainability outcomes. Such results indicate that the manifestation of GDTL traits by leaders can encourage GDM among followers, which in turn complements organizational efforts to orchestrate a successful GDT.

Hypothesis five, which postulated a significant moderating effect of

Table 7
Model compared with and without control variable.

With Control Variable				Without Control Variable			
	Original sample	t values	p values		Original sample	t values	p values
GDM - > GDT	0.351	5.299	0.000	GDM - > GDT	0.358	5.560	0.000
GDTL - > GDM	0.291	3.761	0.000	GDTL - > GDM	0.276	3.612	0.000
GDTL - > GDT	0.206	3.268	0.001	GDTL - > GDT	0.188	3.107	0.002
OGC - > GDM	0.231	2.857	0.004	OGC - > GDM	0.241	2.945	0.003
OGC x GDTL - > GDM	0.193	3.284	0.001	OGC x GDTL - > GDM	0.212	3.517	0.000
OS - > GDM	0.085	1.507	0.132				
OS - > GDT	0.065	1.092	0.275				
Age - > GDM	0.030	0.485	0.628				
Age - > GDT	−0.047	0.808	0.419				

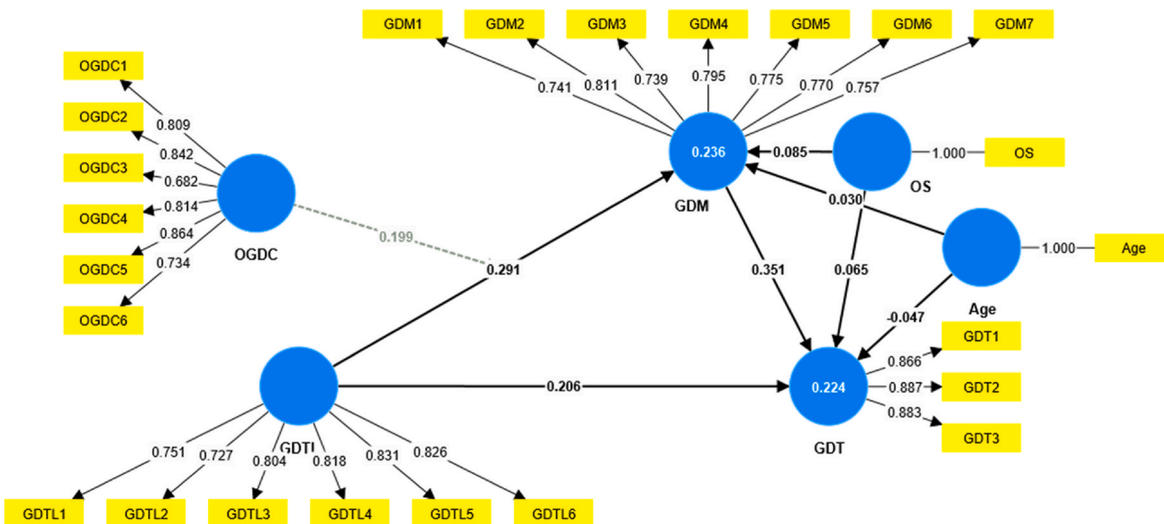

Fig. 6. Model with Control Variable. Age = Participant age and OS = Organization Size.

OGDC in the GDTL-EGDM linkage, has also been empirically supported by the study's findings, which means that the impact of GDTL on GDM is more significant when organizational culture strongly endorses green initiatives and sustainability. This supports the findings of some of the previous studies undertaken in this area, such as Wang et al. (2023) and Xue et al. (2022). By cultivating a sustainable and digitally innovative culture, GDTLs can augment employees' predispositions and preferences to adopt eco-friendly digital tools, technologies, and practices, thereby advancing the materialization of an organization's environmental and digital goals. By cultivating a culture that prioritizes green and digital initiatives, organizations can augment their employees' GDM (Katz et al., 2022; Rosário and Dias, 2022; Xu et al., 2023), thereby facilitating the effective execution of GDT endeavors (Feng et al., 2022; Peng et al., 2021). This finding underscores the significance of effective leadership in promoting environmentally responsible behaviors through fostering the assimilation of digital technologies to achieve ecological sustainability, which is consistent with Sun et al. (2022).

Putting in a nutshell, the results of this study indicate that GDTLs' efforts to bring about a GDT could succeed if they could rally support from their followers through nurturing GDM within a workforce wherein the efficacy of GDTL to affect EGDM increases manifold if a potent OGDC is put in place. Alternatively, cultivating an organizational culture that places importance on environmentally conscious practices can ameliorate the efficacy of GDTL's efforts in culminating positive attitudes towards sustainability and digitalization among employees to actualize a green digital transformation.

5.2. *Theoretical contributions*

The study yields various contributions. Firstly, the findings of this study add to the understanding, integration, and extension of the Stimulus-Organism-Response (SOR) framework and the transformational leadership theory to the GDT context. The study provides significant insight into how external stimuli, such as GDTL and OGDC, influence employees' GDM of employees (Organism) to elicit their positive contributions (response) in the successful orchestration of GDT. Secondly, the study enhances our understanding of the contextual conditions supporting GDT by examining the interaction between GDTL and OGDC in maturing EGDM. The research recognizes the interrelatedness and intricacy of environmental and digital systems following systems thinking and complexity theories. The need to adopt a holistic approach to green digital transformation is highlighted, taking into account the interdependencies and feedback loops among technological, environmental, social, and economic issues. Thirdly, the study contributes significantly by providing interdisciplinary insights and bridging the gap between many academic fields, such as corporate culture, environmental psychology, and leadership studies. The study combines organizational culture and digital transformation theories with environmental sustainability, suggesting that a green digital culture is crucial for achieving successful green digital transformation. It highlights the importance of corporations fostering a culture that prioritizes environmental responsibility and utilizes digital technologies to accomplish sustainability objectives. The understanding of how these factors interact and have an impact on each other is improved by this interdisciplinary approach. Fourth, the academic literature is enriched by this research's depth and breadth of understanding of how organizations negotiate the nexus of sustainability and technology. The study extends transformational leadership theory by stressing the significance of integrating environmental sustainability factors into leadership practices. It highlights how leaders may encourage and persuade followers to adopt green digital transformation through visionary communication, intellectual stimulation, individualized thoughtfulness, and inspirational motivation. Fifth, the newfangled model enhances our understanding of the (behavioral) dynamics of GDT in the unique but neglected Saudi (or Middle Eastern) context. Lastly, the findings of this study could add to the universality of the etic theories advanced from the West to different regional and/or socio-cultural settings.

5.3. Practical implications

The outcomes of this research have several practical insinuations that organizations can consider. Firstly, the findings of this study implicate the organizations in the cultivation and advancement of GDTL as an instrument to spark GDT. The results indicate that leaders who demonstrate GDT behaviors can have a favorable impact on the ecological mindset of their followers to effectuate their positive contributions to GDT. Consequently, organizations should focus on GDTL development to enable digitalization-enabled eco-friendly initiatives and practices through training and development programs, mentoring, practical self-awareness, stakeholder engagement, etc. The sustainability goals and principles should be integrated into the company's mission statement, values and performance objectives. Secondly, an effective OGDC that fosters sustainability and digital innovation must be developed to create an enabling environment for the GDTLs' efforts to produce desired GDT outcomes. This could be effectively done by creating green digital teams, deputing green digital champions, educating followers on green digital initiatives, encouraging their participation in such initiatives, maintaining a continual feedback loop, fostering/cementing green digital readiness at all levels, etc. Workshops and seminars should be organized to educate both the leaders and the employees about the importance of sustainability and the role digital technologies can play in facilitating a green culture. Thirdly, by creating effective configurations with GDTL, OGDC, and EGDM, organizations could profoundly enhance the efficacy of their strategies and efforts to institute a successful GDT. This could be done by educating and enabling leaders to exercise creative leadership configurations contingent upon leader and follower characteristics (e.g., followers' readiness, interpersonal trust, referent power, etc.), organizational characteristics (e.g., complexity, lifecycle, culture, etc.), and situational contingencies (e.g., environmental uncertainty, technological shifts, competition, etc.). Companies can offer mentorship opportunities where experienced leaders can guide employees in integrating sustainability considerations into their activities. Fourthly, organizations must continuously strive to exploit the nexus, synergy, and/or complementarity between green and digital resources to enhance their green digital performance. Key performance indicators and metrics can be established to track the progress related to digital transformation goals. By implementing these pragmatic approaches, entities can establish a sustainable culture, involve their workforce in eco-friendly endeavors, and contribute to a more ecologically sound and sustainable future.

Additionally, courses and programs amalgamating leadership, sustainability, and digital transformation could be contrived based on the findings of our study, which could also serve as an effective subject matter for classes, workshops, and training courses aiming to enable future leaders to meet the challenges of the dynamic marketplace in this Silicon Age. At the public policy level, academic discoveries influence policy proposals. Findings from the research can be used to promote legislation that encourages and supports green digital transitions across a wide range of sectors.

6. Conclusion

This study examined the nexus between GDTL, GDC, GDM, and GDT. The findings reveal a significant positive role of GDTL in affecting GDT directly and indirectly through maturing EGDM, underscoring the significance of leadership conduct in propelling ecological sustainability in corporate settings. Furthermore, a positive mediating role of EGDM in the GDTL-GDT linkage and the moderating role of OGDC in the relationship between GDTL and EGDM has also been empirically substantiated, highlighting the complex dynamics of the successful orchestration of GDT in this knowledge and sustainability-driven age.

We envision a world where sustainability and digital transformation go hand in hand, industrial settings where organizations embrace green leadership, cultivate a culture of environmental consciousness, and create psychological enablements like EGDM, and a society in which individuals are inspired to make sustainable choices in their everyday lives, both within and outside of work. The narrative centers around a central character who exemplifies the tenets of GDTL, guiding a group toward pioneering digital remedies that foster commercial triumph and prioritize environmental welfare. However, significant impediments could surface as this journey progresses. In their individual and team capacities, the employees may encounter resistance, competing priorities, and unforeseen obstacles. However, their unwavering dedication to sustainability could engender persistence in overcoming these obstacles. An effectual exhibition of GDT behaviors from their leaders augmented by a strong OGDC could create just the right enabling environment for these employees to contribute cogently to the GDT initiatives aiming at electrifying their organizations' sustainability performance.

6.1. Limitations and future research directions

Although this study significantly contributes to the comprehension of the association between "green digital transformational leadership" (GDTL), "green digital mindset" (GDM), and" green digital transformation" (GDT), it is essential to acknowledge certain constraints. The constraints underscore domains in which future investigations can augment our comprehension of the subject matter.

First, the generalizability of the results may be constrained by the specific sample and context in which they are based. Subsequent investigations should replicate this study by employing heterogeneous samples from various industries and cultural contexts to augment the generalizability of the results. Comparative studies that examine the manifestation of GDTL, GDM, and GDT across diverse settings, including various industries, organizational sizes, and cultural contexts, would enhance our understanding of these phenomena. Exploring possible discrepancies in the associations between these constructs within different settings can provide insights into the limitations and situational elements that influence these associations. Intervention studies are also valuable for investigating the effectiveness of targeted strategies in promoting sustainability and digital transformation. Such studies may involve implementing interventions facilitated by OGDC to develop GDTL, foster GDM, and promote GDT. Through these interventions, valuable insights can be gained regarding the effectiveness of specific strategies in driving the desired outcomes. These types of studies have the potential to provide valuable insights into evidence-based approaches that organizations can adopt when pursuing sustainability goals.

Second, this research utilized a cross-sectional design, which captures a momentary representation of the associations between variables at a particular instance. Longitudinal or experimental methodologies may offer a more comprehensive understanding of the causal associations among GDTL, EGDM, and GDT over time. This methodology could yield valuable perspectives on the enduring ramifications and interplay of said constructs and their developmental trajectory within organizational contexts.

Third, the research depended on self-report measures, which are susceptible to common method and potential response biases. Subsequent investigations should consider including objective metrics or diverse data sources to augment the veracity of outcomes. A multilevel analysis can provide a wide-ranging understanding of the interplay and impact of GDTL, OGDC, EGDM, and GDT constructs across various organizational levels, including individual, team, and organizational levels. Multilevel analysis can apprehend the subtleties and fluctuations of these interconnections within intricate organizational frameworks.

Fourth, the data utilized in this investigation were obtained from a single source, with a predominant reliance on employees' subjective evaluations and self-appraisals. Integrating multiple data sources, including assessment of leaders and objective performance metrics, can yield a more comprehensive understanding of the relationships being examined. Further investigation is warranted to explore the mediation and moderation mechanisms underlying the association between GDTL, GDM, and GDT. Examining variables such as personal drives, societal expectations, and institutional resources can facilitate the identification of supplementary factors that impact these associations.

Fifth, further investigations may focus on examining the influence of contextual conditions like digital maturity and digital capabilities (at individual and firm levels) in enabling GDTL to produce desired GDT outcomes. Examining such associations could yield valuable information on how firms can successfully incorporate green digital efforts into their broader digital plans. Investigating the relationships between an organization's digital abilities and GDTL, GDC, and GDM could provide insights into the skills and resources required to promote an environmentally conscious digital mindset and carry out green digital transformation projects.

CRediT authorship contribution statement

Mahmoud Abdulhadi Alabdali: Writing – review & editing, Writing – original draft, Project administration, Conceptualization. **Muhammad Zafar Yaqub:** Software, Methodology, Formal analysis, Data curation, Conceptualization. **Reeti Agarwal:** Writing – review & editing, Visualization, Validation, Methodology. **Hind Alofaysan:** Writing – review & editing. **Amiya Kumar Mohapatra:** Writing – review & editing.

Declaration of competing interest

The author(s) declare no conflict of interest with respect to the research, authorship, and/or publication of this article.

Data availability

Data will be made available on request.

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Appendix

Table A1
Measurement Items

Construct	Item	Adopted from
GDTL	1. The leader provides a clear environmental and digital technology vision for the followers to follow.	Chen and Chang (2013)
	2. The leader inspires followers with environmental plans to have impact, utilizing digital technologies and sustainable results.	
	3. The leader gets the employees to work together for the same green environmental, digital, and sustainable goals.	
	4. The leader encourages employees to achieve environmental and digital transformation goals.	
	5. The leader acts by considering the environmental beliefs related to green and digital practices of the individuals.	
	6. The leader stimulates subordinates to think about green and digital ideas and initiatives.	
GDC	1. The firm provides information to employees to understand the importance of sustainability and digitalization.	Marshall et al. (2015)
	2. The firm tries to promote sustainability through digitalization as a major goal across all departments.	
	3. The firm has a clear policy statement urging sustainability and green digital practices in every area of operations.	
	4. Sustainability and green digital practices were a high priority activity in the firm.	
	5. Sustainability and green digital practices were a central corporate value in the firm.	
	6. The firm has a responsibility to be sustainable green digitalized organization using the most recent technology	
GDM	1. I feel personally obligated to save as much energy as possible by using digital technology.	Steg et al. (2005)
	2. I feel morally obligated to save energy such as using smart devices, low-consumption energy machines. etc., regardless of what others do.	
	3. I feel guilty when I waste energy	
	4. I feel morally obliged to use green digital technology instead of regular electricity	
	5. If I would buy a new appliance, I would feel morally obligated to buy a green technology device that has an energy-efficient and recyclability	
	6. I feel obliged to bear green practices, the environment, and nature in mind in my daily behavior	
	7. I would be a better person if I use modern digital technology that saves energy	
GDT	1. Our firm is driving new business processes built on [green] technologies	Aral and Weill (2007)
	2. Our firm is integrating [green] digital technologies into its business activities and operations.	
	3. Our business operations are shifting towards making use of [green] digital technologies	

Table A2
Demographic Items

Region:	2. Specialist or supervisory level
1. Saudi Arabia	3. Manager/Sr Manager level
2. Other GCC	4. Head/Director level
3. Middle East	5. Leadership/C-suite level
4. Europe	Department:
5. North America	1. Sales

(*continued on next page*)

Table A2 (*continued*)

6. South America	2. Marketing
7. Africa	3. Supply chain/logistics
8. Asia	4. HR
Gender:	5. IT
1. Male	6. Strategy
2. Female	7. Finance/accounting
Age:	8. Administration/other support function
1.20–30 years old	9. Operations
2.31–40 years old	Industry:
3.41–50 years old	1. FMCG
4.51–60 years old	2. Healthcare and Pharma
Total Year of Experience:	3. Food and beverage
1.1 year or less	4. Manufacturing
2.2–5 years	
3.6–10 years	5. Oil & gas
4.11–15 years	6. Retails
5.16–20 years	7. IT Technology
6.21 years and above years	8. Telecommunication
Educational Level:	9. Public sector/Government entity
1. High school or less	10. Aviation
2. Diploma	
3. Bachelor	11. Other
4. Master	12. Education
5. PhD	13. Real estate
Occupational level:	14. Banking/investment
1. Entry level	15. Insurance/brokerage
	Organizational size:
	1. Less than 100 employees
	2.100–500 employees
	3.501–1000 employees
	4.1001–5000 employees
	5. More than 5000 employees

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